

Comparative Study of Single and Double Gate All Around Cylindrical FET Structures for High-K Dielectric Materials

Ronit Dutta

PhD researcher, SMT pillar

Sub : NANOELECTRONICS



SINGAPORE UNIVERSITY OF
TECHNOLOGY AND DESIGN



Comparative Study of Single and Double Gate All Around Cylindrical FET Structures for High-K Dielectric Materials

Rajveer Kaur¹ · Balwinder Singh¹

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Abstract

The performance of conventional MOSFET deteriorates because of the short channel effects that appears when it is scaled into nm regime. The existing techniques like channel scaling, variations in the work function implemented on a MOSFET could no longer confront these limitations which demand for the necessity of some upgraded devices and materials that would overcome these shortcomings and offer ameliorate performance. Two experimentally based devices gate all around (GAA) FET and double gate all around (DGAA) FET are modeled and compared at 10 nm and 20 nm respectively. In this paper, the effect of scaling the gate length, oxide thickness and variations in the drain to source voltage utilizing two different gate dielectrics for a single and double GAA cylindrical FET on their respective device performance in terms of drive current (I_{ON}), leakage current (I_{OFF}), switching speed (I_{ON}/I_{OFF}) and subthreshold swing. The study reveals that with a thinner gate oxide, less gate length, less drain to source voltage and with an additional core gate utilizing high-k dielectric materials, the device achieves a better subthreshold slope, higher value of ON-state current, larger ON/OFF current ratio, lesser OFF-state current and lesser power consumption.

Keywords CMOS · MOSFET · Gate all around (GAA) FET · Double gate all around (DGAA) FET · Short channel effects · On current · Off current · Scaling

1 Introduction

Transistor comes into existence in early 1960s plays an essential role in the semiconductor industry. High speed, less area and low power consumptions in the portable devices become the primary concern for researchers in the semiconductor industry [1]. The reduction in the dimensions of a planar CMOS is a cumbersome task. The reduction in the dimensions of a planar CMOS provide better performance and excels the functionality of conventional MOSFET (CMOS) but scaling in nm regimes becomes a cumbersome task. It has an inferior impact on the gate control over the channel which is the main reason for performance degradation of the device as it suffers from short channel effects such as mobility degradation, variations in the threshold voltage (V_{th}), increase in the leakage current and difficulty in increasing the ON-current [1]. According to International

Technology Roadmap for Semiconductor (ITRS) roadmaps (2015) given by Semiconductor Research Corporation, the electronics industry will reach its fundamental limit by 2020s [2, 3]. The number of transistors in a single IC has already reached the maximum scaling limits over last four decades. Scaling of conventional MOSFET demands new technologies, new materials and innovative geometry structures that can ameliorate the gate control over the channel. Multigate geometries like dual gate, tri gate, gate all around geometries are the emerging structures. We approach cylindrical single and double gate all around structures. These structures provide better control of gate over the channel. Cylindrical GAA provides superior performance than single, double and triple gate geometries [5].

The basic difference between MOSFET and GAA FET is the respective geometry [4–7]. In planar MOSFET, the channel is horizontal, whereas the channel of GAA structure is cylindrical. In this, the gate surrounds the entire channel which improves its ability to control the flow of current; it lowers down the corner effects, reduces short channel effects

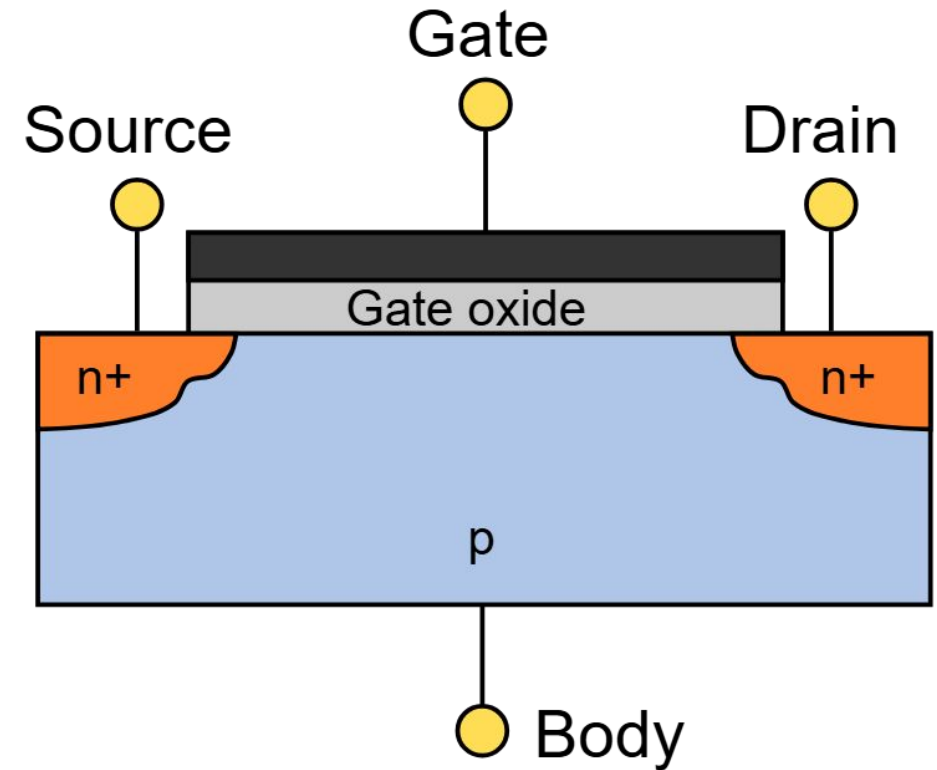
Balwinder Singh
Balwinder@cdac.in

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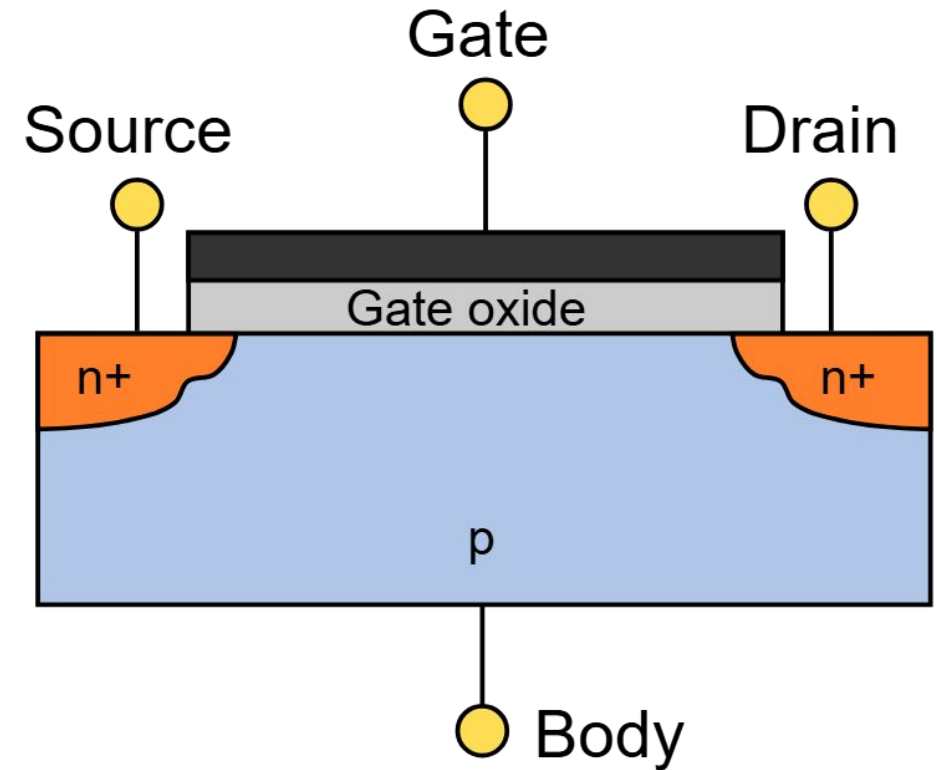
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1. Introduction
2. Short Channel Effects
3. Evolution of FET geometries & associated physics.
4. Simulation and parameter tuning analysis
5. Performance metrics and Fabrication issues
6. Future directions



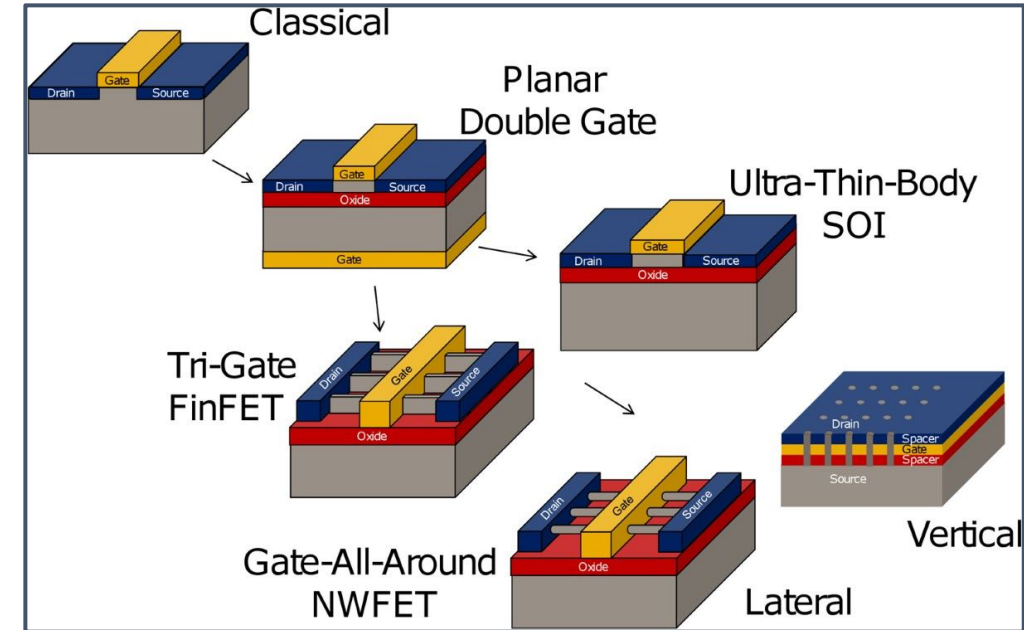
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Introduction

1. CMOS scaling into the nanometer regime breaks traditional MOSFET behavior and conventional tricks such as shorter channels, work-function tweaks no longer fix the degradation.
2. To regain gate control and reduce leakage, new device geometries like Fin, Gate-All-Around (GAA) and Double-Gate-All-Around (DGAA) have emerged.
3. This paper evaluates **cylindrical GAA** and **cylindrical DGAA** structures as theoretical candidates for sub-20 nm nodes.
4. The study systematically varies gate length, oxide thickness, drain-bias, and dielectric material to understand performance trends.



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Power Dissipation

	Static Power Loss	Dynamic Power Loss
Primary source	Dominated by leakage currents: subthreshold leakage, DIBL-induced leakage, GIDL.	$P_{dyn} = \alpha C_{gate} V_{DD}^2 f$
Electrostatics	Weak electrostatics (planar, thin SiO ₂) allow drain field to penetrate channel → high I _{OFF} → static power ↑.	Poor geometries require higher charge to switch channel → higher effective C _{gate} .
Geometry dependence	FinFET → GAA → DGAA progressively tighten gate control, sharply reducing leakage-driven static power.	Stronger geometries (GAA/DGAA) improve inversion efficiency → lower switching charge for same drive, reducing dynamic power despite higher Cox.

Velocity saturation

1. CMOS scaling into the nanometer regime breaks traditional MOSFET behavior. Shorter channels produce higher lateral electric fields.
2. When carriers reach critical vertical field, velocity saturates which means that current no longer scales with gate field. This causes a loss of transconductance, degraded drive current, and poor switching efficiency.

Gate voltage controls charge

$$Q_{inv} \propto (V_G - V_T)$$

But current is also controlled by velocity: $I_D = Q_{inv} \cdot v$

At low lateral field (linear region):

$$v = \mu E_{lat}$$

BUT: at high lateral field (near drain):

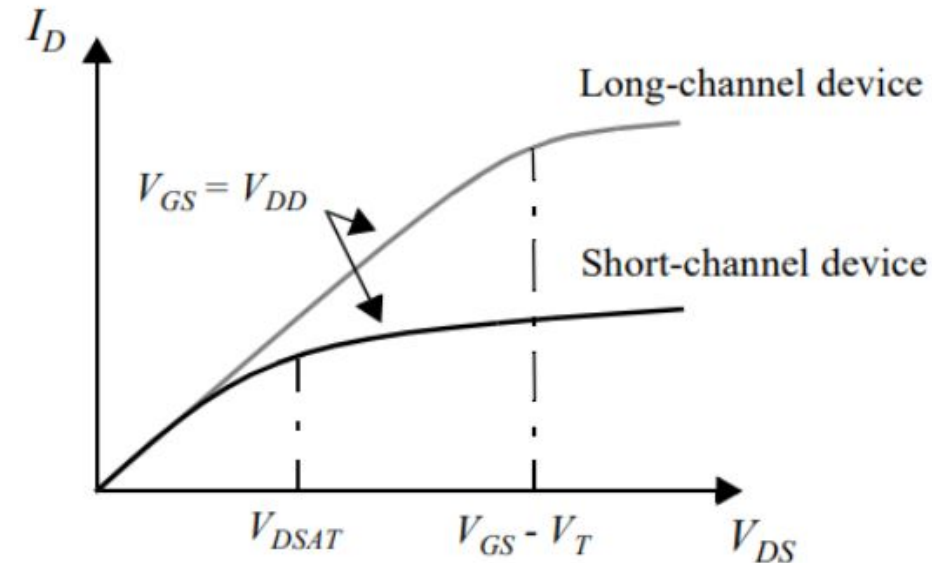
$$v \rightarrow v_{sat}$$

Carrier velocity cannot increase anymore — it hits a **physical limit**.

So now:

$$I_D \approx Q_{inv} \cdot v_{sat}$$

Even if you pump V_G higher, v **stays fixed**, so current barely increases.



Short-channel devices display an extended saturation region due to velocity-saturation.

Hot Carrier Injection and GIDL (Gate Induced Drain Leakage)

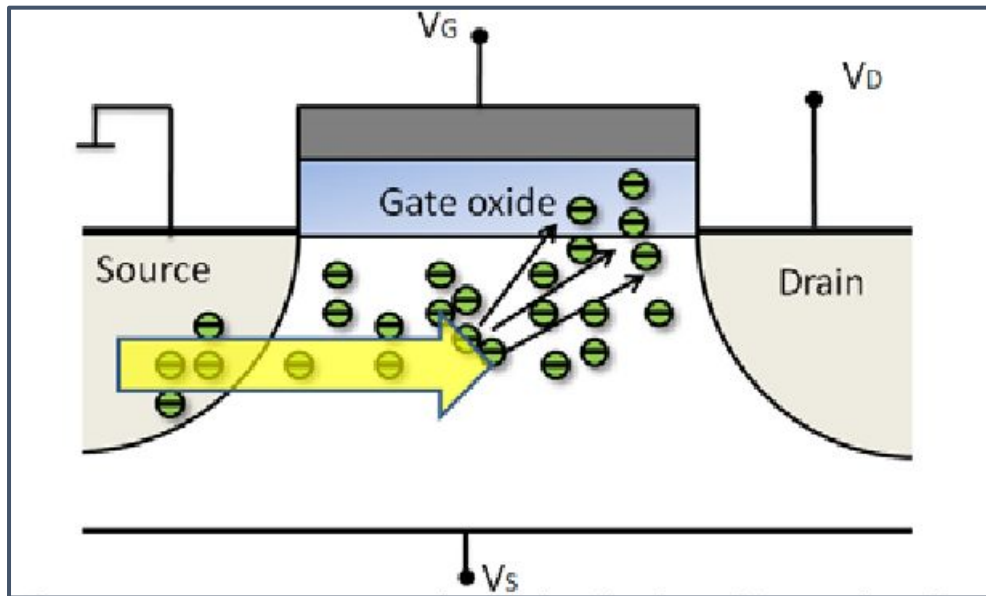


Fig : Hot Carrier Injection

- **HCI** : Short channels create high lateral field causing carriers to accelerate and pick up excess kinetic energy.
- These hot carriers gain enough energy to jump into or damage the gate oxide.
- This creates interface traps and oxide charge causing permanent device degradation.
- Injected charge shifts threshold voltage and reduces mobility.
- HCI accumulates over billions of switching cycles.
- This leads to aging, transconductance loss, and eventual device death under high-field stress.

- **GIDL** : As the oxide gets thinner and drain voltage increases, a very high electric field forms near the gate–drain edge.
- This strong field bends the bands steeply, enabling electrons to tunnel from the valence band to the conduction band giving leakage even when the device is OFF.
- Persistent GIDL contributes to oxide damage, worsens static power, and limits how low standby power can go.

DIBL (Drain Induced Barrier Lowering) & Punch Through Effect

- **DIBL** : The source–channel barrier is reduced, effectively lowering Threshold voltage at high drain voltage.
- With the source barrier lowered, electrons enter the channel even when Gate voltage is below threshold. $I(\text{off})$ increases.
- Device gets “mode coupled” to drain voltage instead of gate voltage, which creates unpredictable behaviour.

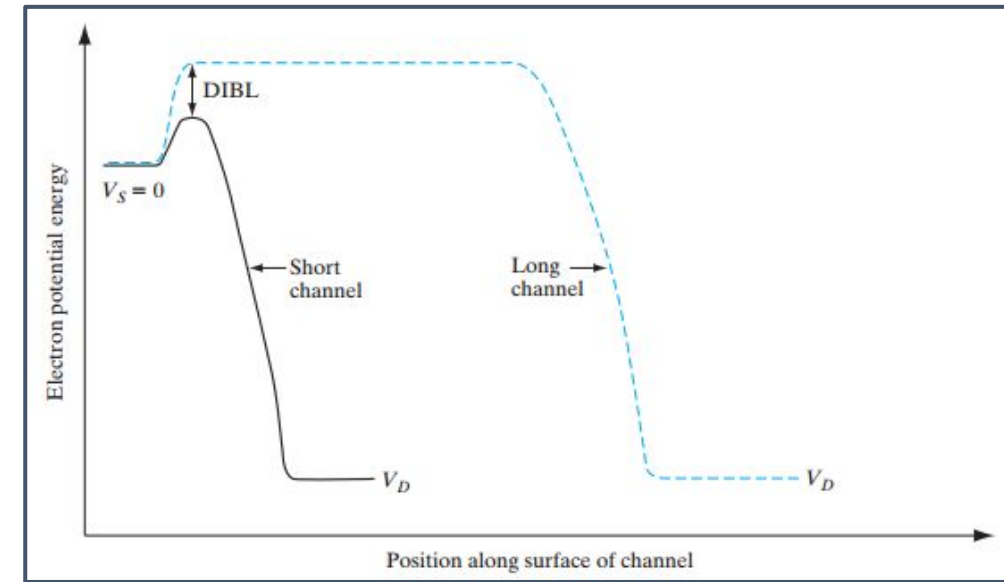


Fig : DIBL

- **Punch-Through Effect** : As channel length shrinks, the source and drain depletion regions extend deeper into the silicon body.
- When these regions touch, the device forms a weakly conductive path between S and D even below cutoff and subthreshold causing a steep rise in drain current in the OFF state and overpowering the gate's ability to shut the transistor off.
- Once punch-through begins, V_t collapses, device isolation is compromised.

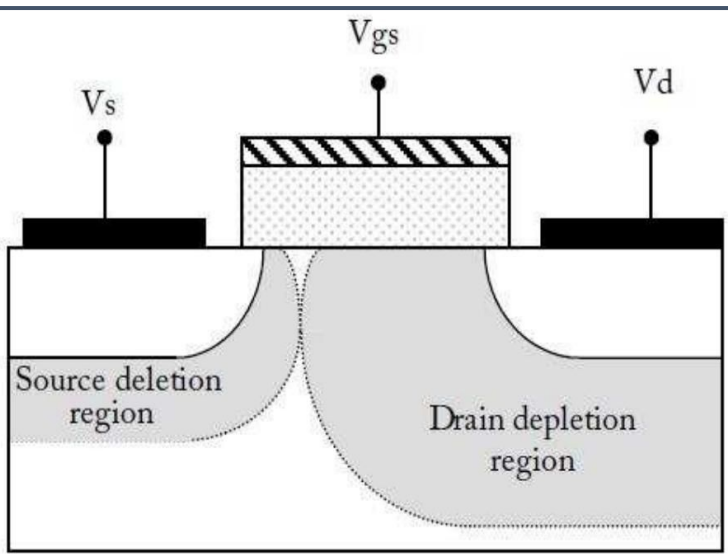
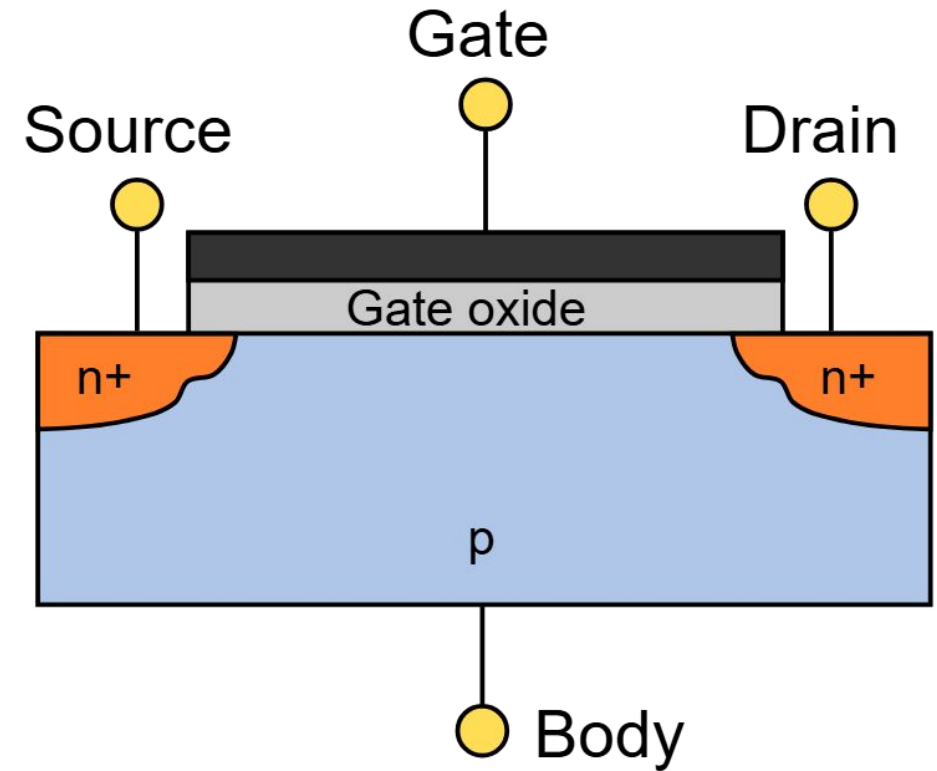


Fig : Punch Through Effect

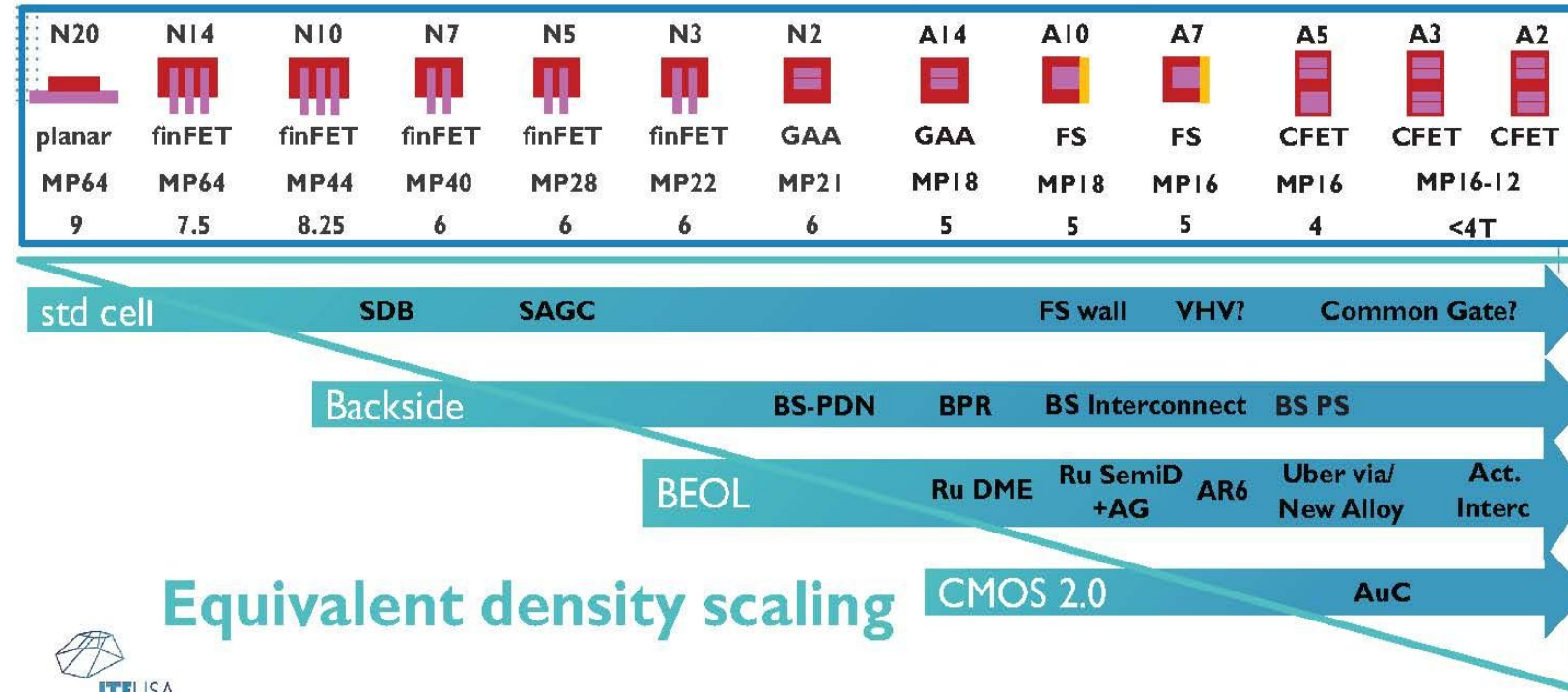
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FET Tech Evolution and requirement

Transistor density scaling



- Gate control improves when the C_{ox} dominates C_{dep} . This makes sure that voltage drop is minimal in the oxide layer and mostly dropped by the depletion capacitance.
- Increasing the physical area over which the gate touches the channel directly increases C_{ox} giving more electrostatic control.

Equivalent density scaling



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- Larger controlled surface leads to more uniform inversion charge distribution which in turn leads to higher drive current.
- Scaling demands that as L shrinks, gate control must increase faster than drain influence and therefore larger interface area is the only geometry-based way to achieve this.

FinFETs

1. FinFETs were the first candidate in structural improvements. Gate wraps around three surfaces of a thin silicon “**fin**”.
2. Fin thickness shrinks which leads to depletion region being constrained hence better barrier control.
3. Due to the fin the effective width $W = (2H+W)$. Fin design transforms MOSFET from a 1-sided gate to a tri-gate system. This improves electrostatics without drastically changing process flow.
4. In terms of SCE :
 - a. Reduced DIBL: drain field has fewer “escape paths” into the channel.
 - b. Reduced punch-through: thin fin body prevents depletion merging.
 - c. Better subthreshold behavior: increased C_{ox}/C_{dep} .

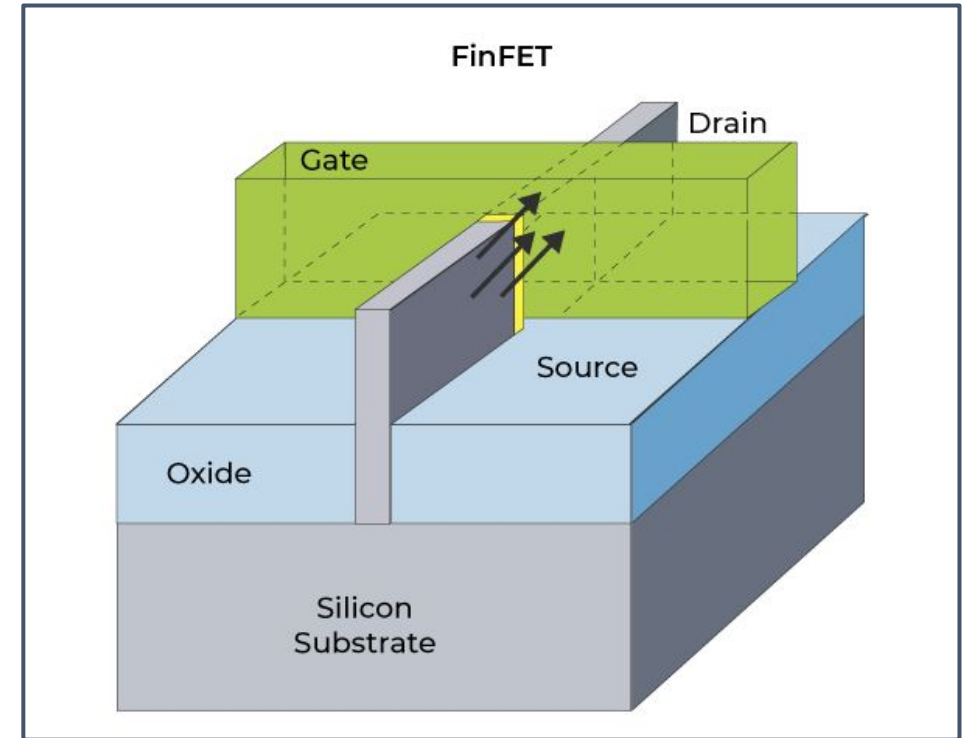


Fig : FinFET Design (**crude**)

GAAFETs

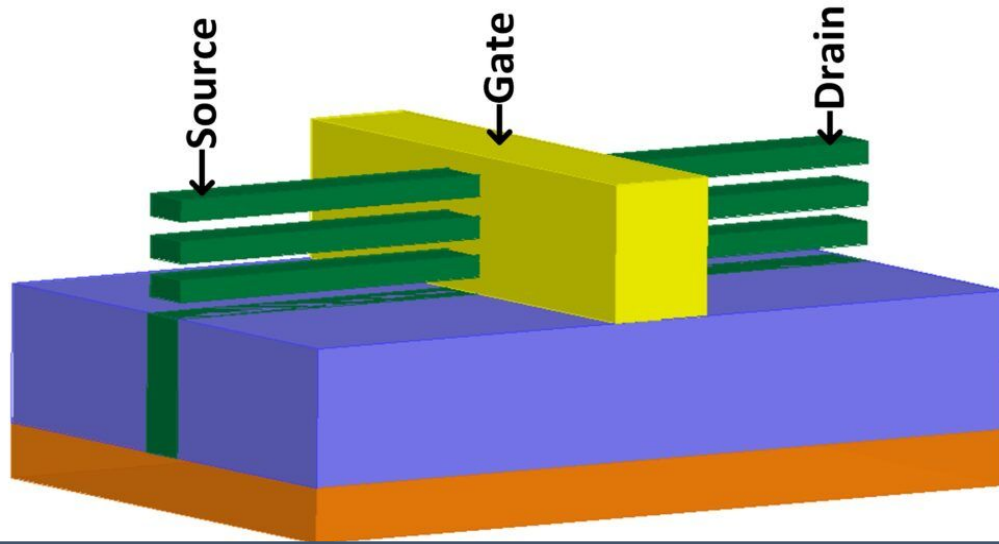
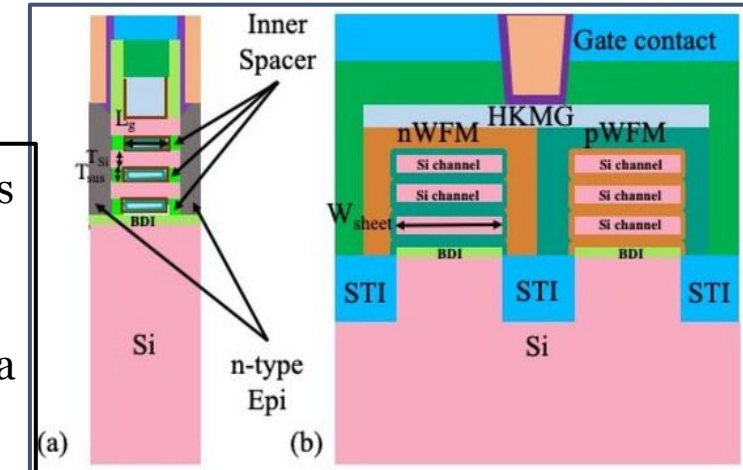


Fig : GAA-FET design (**Better**)

- In GAA FETS the gate surrounds the channel on all four sides. The geometry is rectangular. Channel cross-section becomes a nanosheet or nanowire.



- Because of 4 sided contact gate-channel coupling is stronger which leads to stronger SCE suppression.
 - a. Full wrap means no uncontrolled surfaces left for drain field penetration.
 - b. Lower leakage and tighter electrostatic confinement therefore lower subthreshold conduction and barrier lowering.

Fig : GAA-FET design (**crude**)

- GAAFET eliminates the last uncontrolled face of the fin — the gate now captures the channel completely!`

Cylindrical GAA/ DGAA

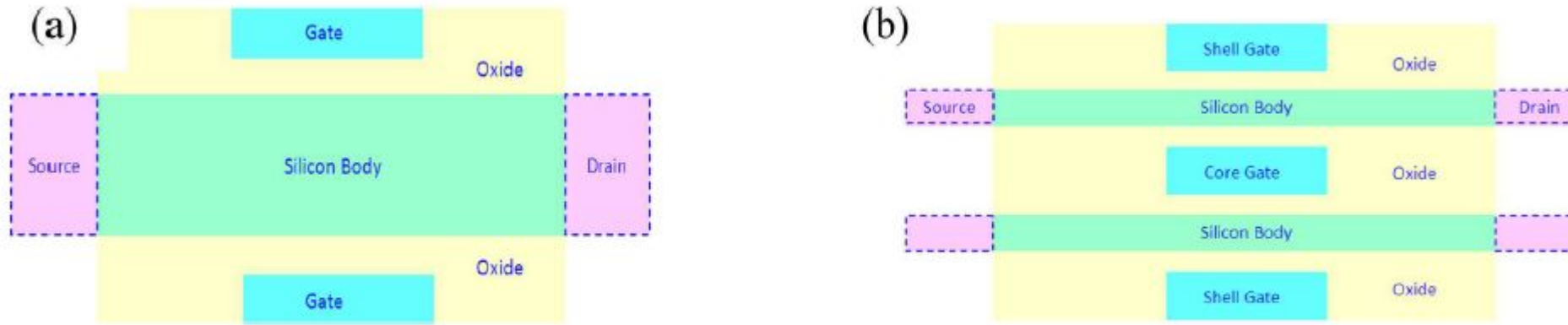


Fig. 2 Cross-section view of the simulated **a** GAA FET and **b** DGAA FET

This paper gives structures for cylindrical geometry for GAAFETs...

1. **CGAA** : Radial symmetry eliminates corners and produces uniform electric field.
 - Strongest possible gate-induced inversion for a single gate.
 - Smallest electrostatic length λ , best punch-through immunity.
2. **CDGAA** : Two independent wrapping gates $\rightarrow$ sum of two capacitive controls: $C(ox, DGAA) = C(inner) + C(outer)$
 - Drain field almost completely screened $\rightarrow$ lowest DIBL.
 - Inversion charge can be modulated more strongly $\rightarrow$ highest $I(ON)$.
 - Best suppression of subthreshold leakage across all geometries.

Why is CDGAA so good?

This is essentially the physically optimal geometry for electrostatics giving :

1. maximum surface contact
2. maximum symmetry
3. maximum gate control
4. minimal influence of drain.

DGAA is the logical end-point of “increasing controlled area” — nothing wraps the channel more effectively.

Why High-K needed?? How does it improve CDGAA?

- The dual-gate geometry dramatically strengthens electrostatic control, because the channel potential is simultaneously regulated from the inner and outer radial boundaries, leaving very little room for drain-induced perturbation. $C_{\text{ox,eff}} = C_{\text{in}} + C_{\text{out}}$.

$$C_{\text{in}} = \frac{2\pi\epsilon_{\text{ox,in}}}{\ln\left(\frac{r_{\text{ch}}}{r_{\text{core}}}\right)}, \quad C_{\text{out}} = \frac{2\pi\epsilon_{\text{ox,out}}}{\ln\left(\frac{r_{\text{shell}}}{r_{\text{ch}}}\right)}$$

- Cylindrical symmetry ensures that the inversion charge forms uniformly around the channel, eliminating corner effects and weak conduction paths that typically degrade control in rectangular or fin-type structures. Due to increased $W(\text{eff})$, transconductance increases (more $I(\text{on})$) and subthreshold swing decreases.

$$W_{\text{eff}} = 2\pi r_{\text{ch}}$$

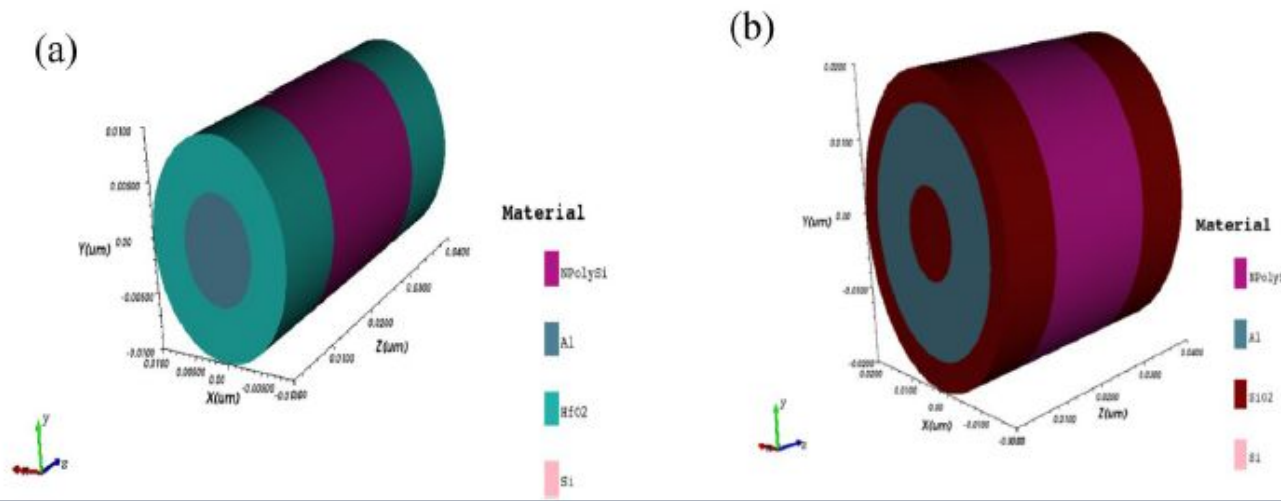
- The combined oxide capacitance of the inner and outer gates effectively adds together, resulting in a substantially larger gate-to-channel coupling that enhances both threshold stability and charge modulation efficiency.

- SiO_2 $k = 3.9 \rightarrow$ tiny C_{ox} unless t_{ox} is dangerously thin
- HfO_2 $k \approx 25 \rightarrow$ huge C_{ox} *without* atomically thin t_{ox}
- High-k benefits:
 - Lower gate leakage
 - Lower E_{ox} stress
 - Higher inversion charge ($I_{\text{on}} \uparrow$)
 - Better SS
- Key point: physics requires **large C_{ox}** , not **thin SiO_2**

- High-k dielectrics increase the achievable gate capacitance without requiring physically thin oxide layers, thereby reducing tunneling leakage and improving gate reliability while still enabling strong gate authority.

Why High-K needed??

Fig. 1 Schematics of the simulated **a** GAA FET with HfO_2 gate dielectric and **b** DGAA FET with SiO_2 gate dielectric

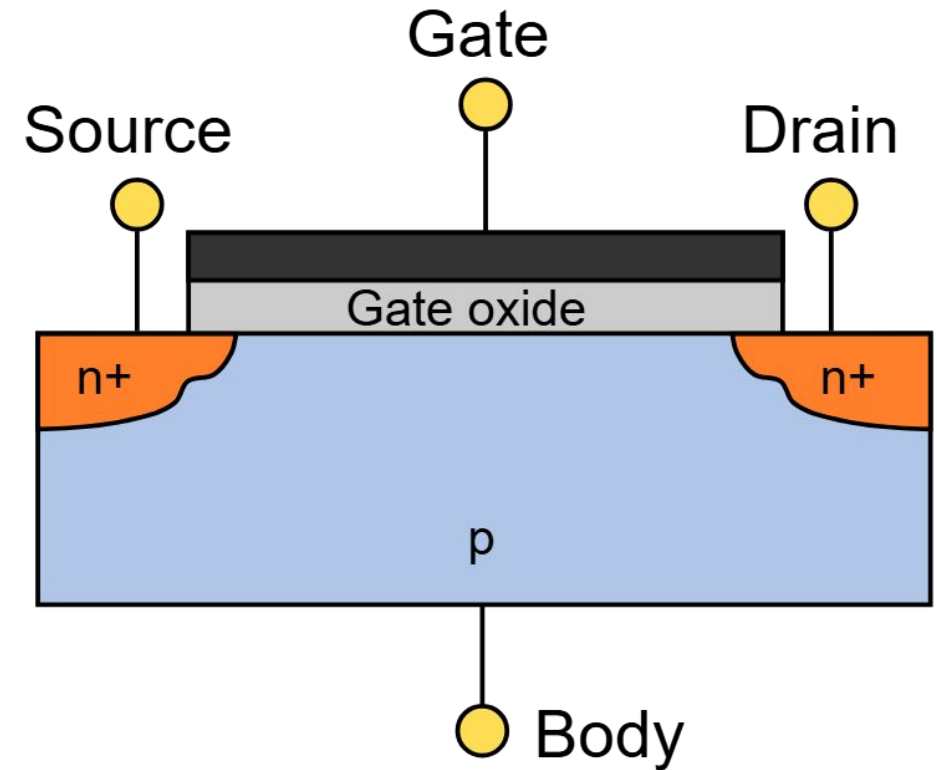


- Short-channel effects such as DIBL, punch-through, and subthreshold leakage are minimized, because the reinforced electrostatics restrict the spatial extent of depletion and prevent the drain field from influencing the source barrier.
- Drive current improves due to higher inversion charge density and reduced electrostatic degradation, allowing the device to sustain strong current even at aggressively scaled gate lengths.

- The overall device becomes less sensitive to variations in channel dimensions and applied drain voltage, since the stronger radially symmetric confinement stabilizes the channel potential under a wide range of bias conditions.
- This architecture represents a near-optimal electrostatic configuration for nanoscale MOSFETs, as it maximizes gate influence while minimizing parasitic effects and physical limitations inherent to planar, FinFET, or single-gate GAA devices.

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Simulation Init Params

Table 1 Parameters used for device simulation

Parameters	CGAA (cylindrical gate all around)			
	SiO2 (Silicon dioxide)		HfO2 (Haf-nium oxide)	
Gate oxide Material				
Technology	20 nm	10 nm	20 nm	10 nm
$V_{DD, lin}[V]$	0.05	0.05	0.05	0.05
$V_{DD, sat}[V]$	0.08	0.08	0.08	0.08
L_G [nm]	20	10	20	10
Radius [nm]	10	10	10	10
Gate work function (eV)	4.45	4.45	4.45	4.45
Donor ion concentration	1E20	1E20	1E20	1E20
Acceptor ion concentration	1E18	1E18	1E18	1E18

Table 2 Parameters used for device simulation

Parameters	CDGAA (cylindrical double gate all around)			
	SiO2 (Silicon dioxide)		HfO2 (Haf-nium oxide)	
Gate oxide Material				
Technology	20 nm	10 nm	20 nm	10 nm
$V_{DD, lin}[V]$	0.05	0.05	0.05	0.05
$V_{DD, sat}[V]$	0.08	0.08	0.08	0.08
L_G [nm]	20	10	20	10
Radius [nm]	20	20	20	20
Gate work function (eV)	4.7	4.7	4.7	4.7
Donor ion concentration	1E20	1E20	1E20	1E20
Acceptor ion concentration	1E15	1E15	1E15	1E15

Gate length scaling

Table 3 Figure of merit for GAA and DGAA FET with SiO₂ as gate dielectric at 10 nm and 20 nm

Parameters	GAA FET (SiO ₂)		DGAA FET(SiO ₂)	
	10 nm	20 nm	10 nm	20 nm
I _{ON}	2.38E-05	1.25E-05	0.0001244	9.58E-05
I _{OFF}	9.56E-08	1.41E-09	2.03E-05	2.91E-07
I _{ON} /I _{OFF}	0.02E+03	0.88E+04	5.40E+06	3.29E+02
SS	149	100	210	189
P _(LEAKAGE)	4.78E-09	7.05E-11	1.62E-07	1.45E-08
P _(CONSUMED)	1.19E-06	6.23E-07	9.45E-06	4.79E-06

Table 4 Figure of merit for GAA and DGAA FET with HfO₂ as gate dielectric at 10 nm and 20 nm

Parameters	GAA FET (HfO ₂)		DGAA FET (HfO ₂)	
	10 nm	20 nm	10 nm	20 nm
I _{ON}	4.80E-05	2.38E-05	0.0001244	0.0001535
I _{OFF}	6.48E-09	4.90E-09	2.03E-05	2.69E-08
I _{ON} /I _{OFF}	0.74E+04	0.48E+05	5.40E+06	5.70E+08
SS	80	70	210	93
P _(LEAKAGE)	1.59E-10	2.45E-11	1.62E-07	1.34E-09
P _(CONSUMED)	2.84E-06	1.19E-06	9.45E-06	7.68E-06

- When the gate length is reduced from 20 nm to 10 nm (fixed oxide and drain bias), the gate gets tighter electrostatic grip over the channel, so the drain current increases for both GAA and DGAA devices.
- Among all cases, the highest drain current appears for the DGAA device with HfO₂ at 10 nm, because the extra core gate plus high-k dielectric pull much more inversion charge into the cylindrical channel.
- Shorter gate length improves drive current but also increases leakage power and makes subthreshold swing worse in DGAA at 10 nm, meaning more gate voltage is needed to significantly change the drain current.
- The paper explicitly notes that the additional gate gives better control over short-channel effects, and HfO₂ can be used instead of SiO₂ to cut leakage power even when the device is aggressively scaled.

Oxide Thickness [t_{ox}] Scaling

- When the oxide thickness is scaled down from 2 nm to 1 nm at 10 nm gate length, the gate couples more strongly to the channel, so the drive current increases for both single and double GAA structures.
- At 1 nm oxide, the DGAA device again shows the maximum drain current, because of the combined effect of dual gates and stronger gate-channel coupling.
- Thinner oxide in this study reduces leakage power, because the improved electrostatics let the device reach the same inversion level at lower effective field stress.
- Using HfO_2 instead of SiO_2 further boosts drive current and lowers both leakage current and leakage power, since high-k allows a large effective capacitance without using an ultra-thin physical oxide. The authors conclude that combining oxide scaling, high-k dielectric, and an additional core gate helps keep short-channel effects under control while still increasing drive current.

Table 6 Figure of merit for GAA and DGAA FET with HfO_2 as gate dielectric for $T_{ox} = 1$ nm and 2 nm

Parameters	GAA FET (HfO_2)		DGAA FET(HfO_2)	
	$T_{ox} = 1$ nm	$T_{ox} = 2$ nm	$T_{ox} = 1$ nm	$T_{ox} = 2$ nm
I_{ON}	2.72E-05	2.38E-05	0.0003261	0.0003178
I_{OFF}	4.25E-10	4.90E-10	2.87E-07	9.79E-07
I_{ON}/I_{OFF}	0.64E+05	0.48E+05	1.11E+05	3.24E+02
SS	149	80	120	161
$P_{(LEAKAGE)}$	1.59E-10	3.24E-10	1.44E-10	4.89E-08
$P_{(CONSUMED)}$	2.84E-06	2.40E-06	1.63E-05	1.59E-05

Table 5 Figure of merit for GAA and DGAA FET with SiO_2 as gate dielectric for $T_{ox} = 1$ nm and 2 nm

Parameters	GAA FET (SiO_2)		DGAA FET(SiO_2)	
	$T_{ox} = 1$ nm	$T_{ox} = 2$ nm	$T_{ox} = 1$ nm	$T_{ox} = 2$ nm
I_{ON}	1.56E-05	1.25E-05	0.0001756	0.000124
I_{OFF}	7.01E-10	1.41E-09	1.50E-07	2.03E-06
I_{ON}/I_{OFF}	0.22E+05	0.88E+04	1.71E+03	6.12E+01
SS	120	70	190	210
$P_{(LEAKAGE)}$	1.02E-09	4.78E-09	7.25E-09	1.62E-10
$P_{(CONSUMED)}$	1.54E-06	1.19E-06	8.78E-06	9.45E-06

V(DS) Scaling on Drain Current

Table 7 Figure of merit for GAA and DGAA FET with SiO₂ as gate dielectric architecture at 0.05 V and 0.08 V

Parameters	GAA FET (SiO ₂)		DGAA FET(SiO ₂)	
	0.05 V	0.08 V	0.05 V	0.08 V
I _{ON}	2.38E-05	3.34E-05	0.0001244	0.0001109
I _{OFF}	9.56E-08	1.39E-07	2.03E-06	1.26E-06
I _{ON} /I _{OFF}	0.29E+03	2.40E+02	6.12E+06	0.0000885E+06
SS	49	90	210	208
P _(LEAKAGE)	4.78E-09	1.11E-08	1.62E-07	6.31E-08
P _(CONSUMED)	1.19E-06	2.67E-06	9.45E-06	5.5E-06

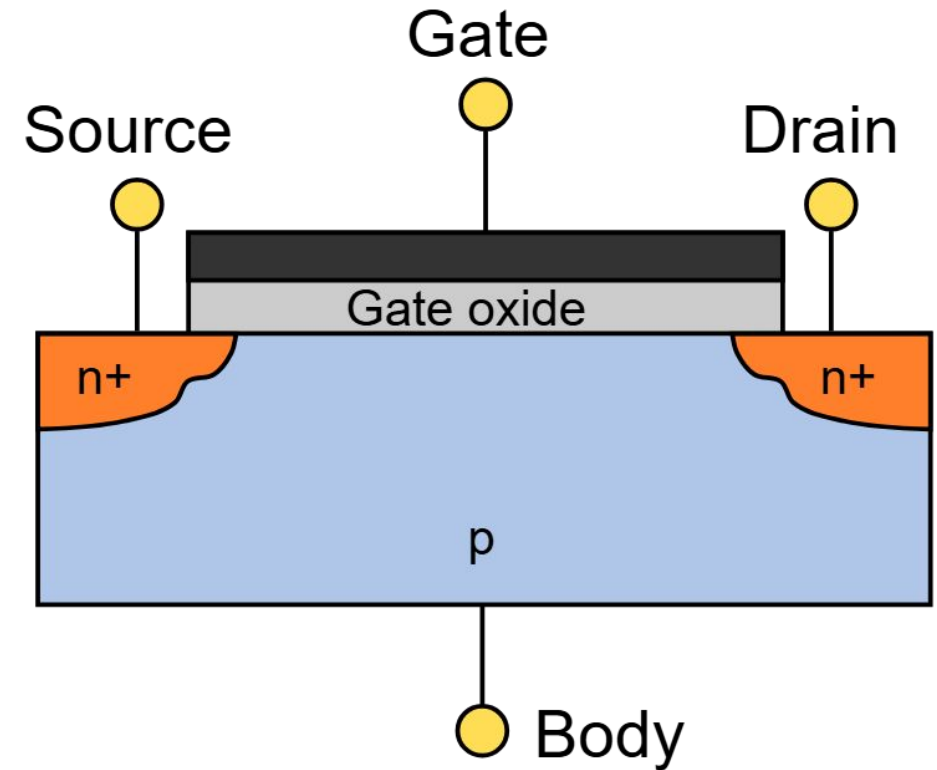
Table 8 Figure of merit for GAA and DGAA FET with HfO₂ as gate dielectric architecture at 0.05 V and 0.08 V

Parameters	GAA FET(HfO ₂)		DGAA FET(HfO ₂)	
	0.05 V	0.08 V	0.05 V	0.08 V
I _{ON}	4.80E-05	7.28E-05	0.0003178	0.0004569
I _{OFF}	6.48E-09	8.20E-09	9.79E-07	1.44E-06
I _{ON} /I _{OFF}	0.74E+04	0.88E+04	3.24E+07	3.17E+02
SS	80	169	161	212
P _(LEAKAGE)	3.24E-10	6.56E-10	4.85E-08	1.15E-07
P _(CONSUMED)	2.40E-06	5.82E-06	1.59E-05	3.66E-05

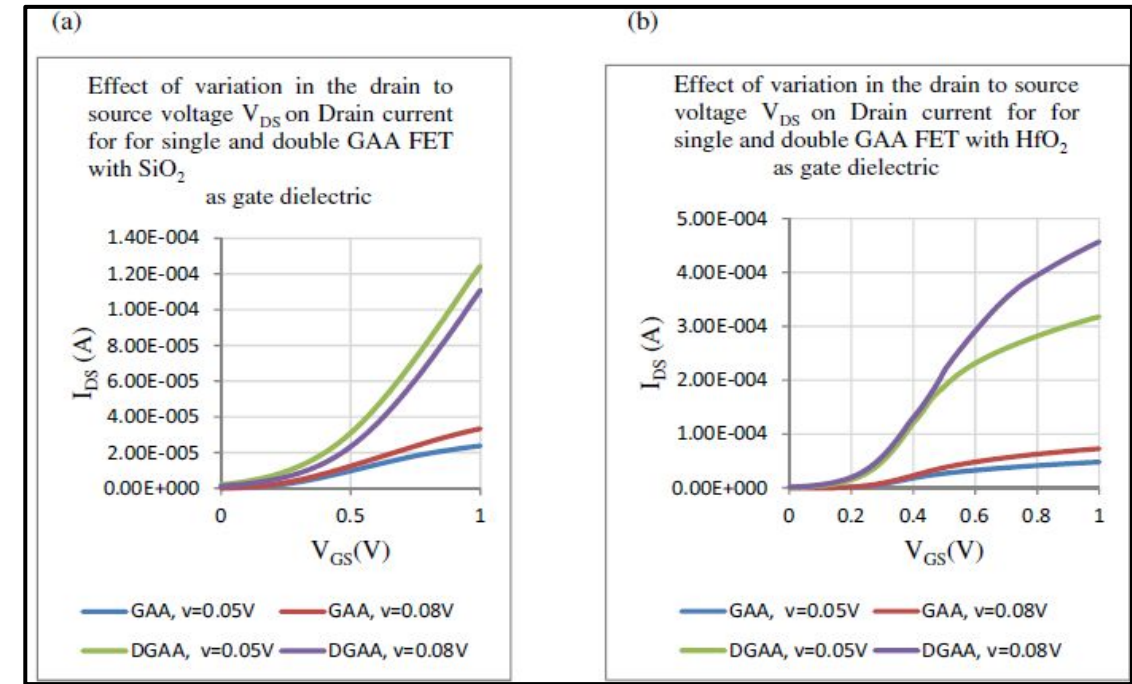
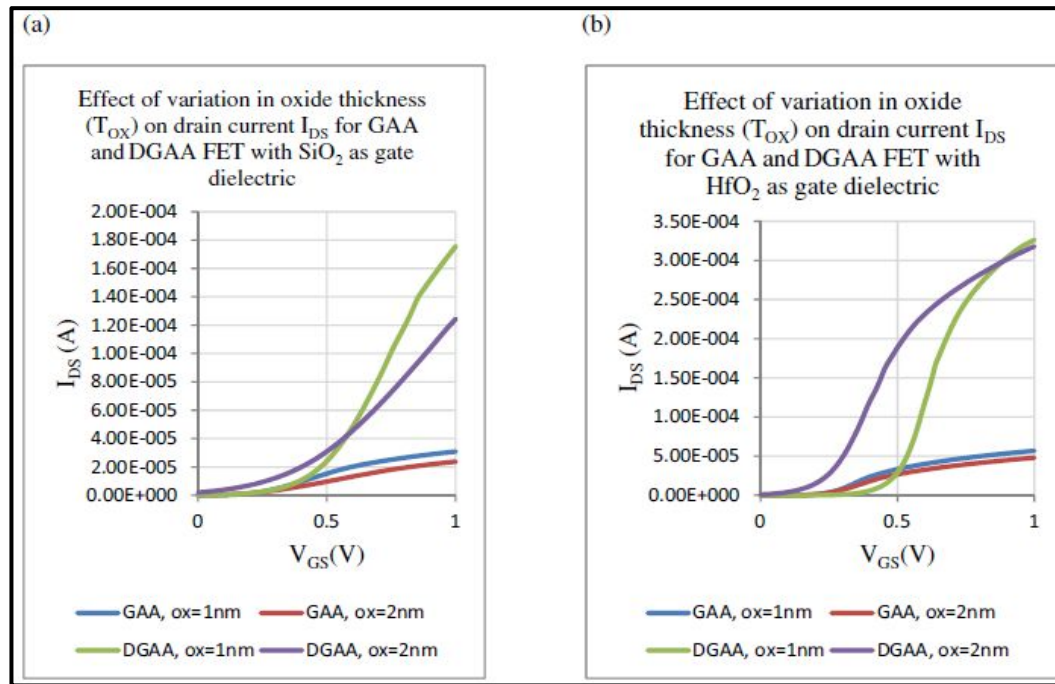
- Increasing the drain–source voltage from 0.05 V to 0.08 V at fixed gate length and oxide thickness naturally raises the drain current in both single and double GAA devices, because the lateral field along the channel becomes stronger.
- For each bias condition, the DGAA structure delivers higher current than the single-gate GAA, showing that its stronger electrostatics translate into more carriers being driven across the channel.
- Across the same drain-voltage sweep, HfO₂-based devices outperform SiO₂-based ones, providing higher drive current while still keeping leakage lower.
- The takeaway from this section is that DGAA with high-k dielectric not only scales well in length and oxide thickness, but also remains robust as drain bias is increased, giving higher current without as severe a penalty in leakage and short-channel behavior.

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Performance Metrics



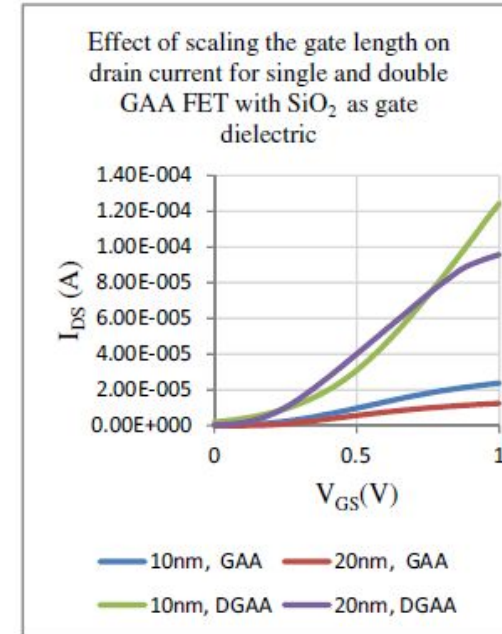
1. Across all gate lengths and oxide thicknesses, DGAA devices consistently show higher ON current than single-gate GAA structures, confirming that the added core gate gives stronger modulation of the cylindrical channel.
2. **HfO₂-based devices** outperform SiO₂ devices in every configuration, with higher ON current and lower OFF current, because the high-k dielectric boosts gate coupling without causing tunneling leakage.

Performance Metrics

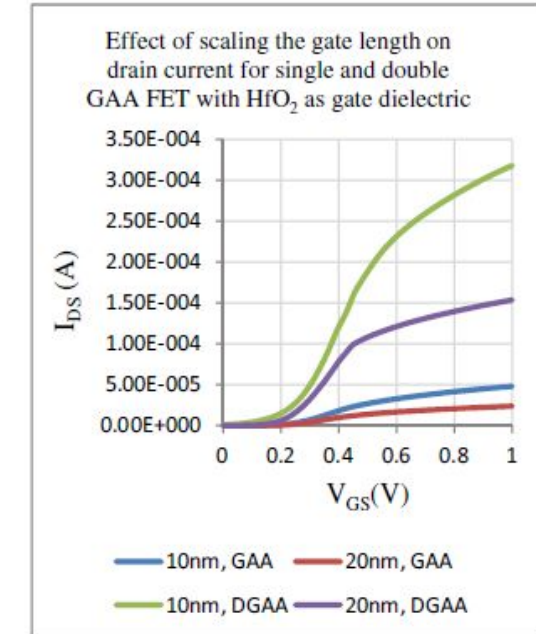
1. ON/OFF current ratio is maximized in the DGAA + HfO₂ combination, especially at 10 nm and 1 nm oxide thickness, where the electrostatics become extremely tight.
2. The paper shows that leakage power decreases in high-k devices due to better control over the barrier at low gate voltages, even when dimensions are aggressively scaled.
3. Although DGAA at 10 nm has slightly poorer subthreshold swing, the authors highlight that its ON current improvement outweighs that limitation, making it the best overall performer.

Fig. 3 Effect of scaling the gate length on drain current for GAA and DGAA FET with **a** SiO₂ as gate oxide and **b** HfO₂ as gate oxide

(a)



(b)



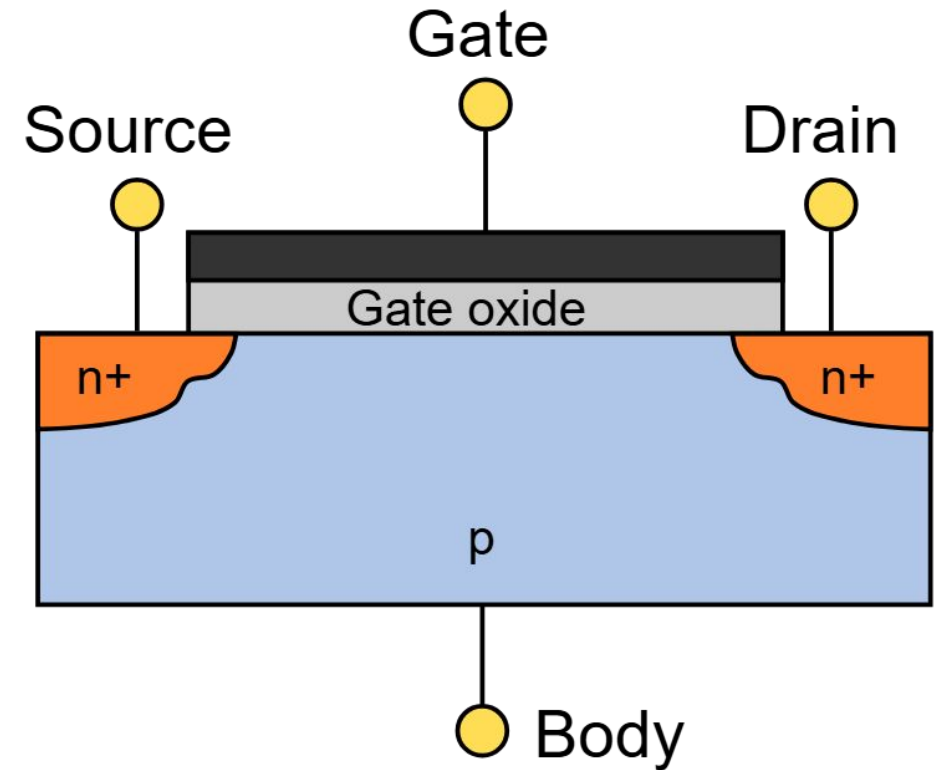
If CGAA and CDGAA is so good then why aren't they ubiquitous?

Obvious question : *“If cylindrical geometry is so obviously better then why isn't it industry standard and omnipresent? Are scientists dumb?”* The following issues come up :

- **Extremely Difficult Shape to Fabricate at Scale :** A perfect nanowire cylinder requires atomically uniform etching and deposition around a 360-degree surface; today's EUV + plasma etch tools naturally produce rectangular features, not circles.
- **Yield Collapses with 3-D Curved Structures :** Billions of identical cylinders on a 300 mm wafer demand nanometer-precision; even tiny variations in diameter, roughness, or alignment cause leakage, variability, and device failures.
- **Gate Stack Deposition Is Harder on Cylinders :** Conformal ALD of high-k + metal around a curved wire is far more complex than depositing on the flat surfaces of nanosheets; defects multiply and reliability becomes unpredictable.
- **Integration With Source/Drain and Contacts Is Non-Trivial :** Connecting raised or recessed source/drain regions to a suspended cylindrical wire without breaking it, deforming it, or creating massive parasitics is a major engineering challenge.
- **Cost and Process Flow Explosion :** Moving from FinFET → nanosheet GAA uses evolutionary process modules; cylindrical DGAA would require new etch chemistries, new metrology, new gate stacks, new mechanical supports, and new reliability rules — meaning billions of dollars and unproven yield.

OUTLINE

1. Introduction
2. Short Channel Effects
3. Evolution of FET geometries & associated physics.
4. Simulation and parameter tuning analysis
5. Performance metrics and Fabrication issues
6. Future directions



Future Directions

1. The results suggest that future nanoscale transistors require more than just lateral scaling; they need stronger 3-D electrostatic confinement, which multigate cylindrical structures provide.
2. Combining high-k dielectric materials with advanced multi-gate geometries is shown to reduce leakage power and enhance drive current, addressing the power-performance trade-offs of deep scaling.
3. The paper demonstrates that DGAA structures remain stable under aggressive variations in gate length, oxide thickness, and drain voltage, indicating robustness needed for sub-10 nm logic nodes.
4. The study positions DGAA as a strong candidate for overcoming the electrostatic limitations faced by FinFET and single-GAA devices at extreme scaling regimes.
5. Because it maintains high ON current and low OFF current simultaneously, DGAA offers a pathway toward energy-efficient high-density ICs as device dimensions continue shrinking.